

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

CO2_AT1 – Chart Construction Test

Integrated Disciplines

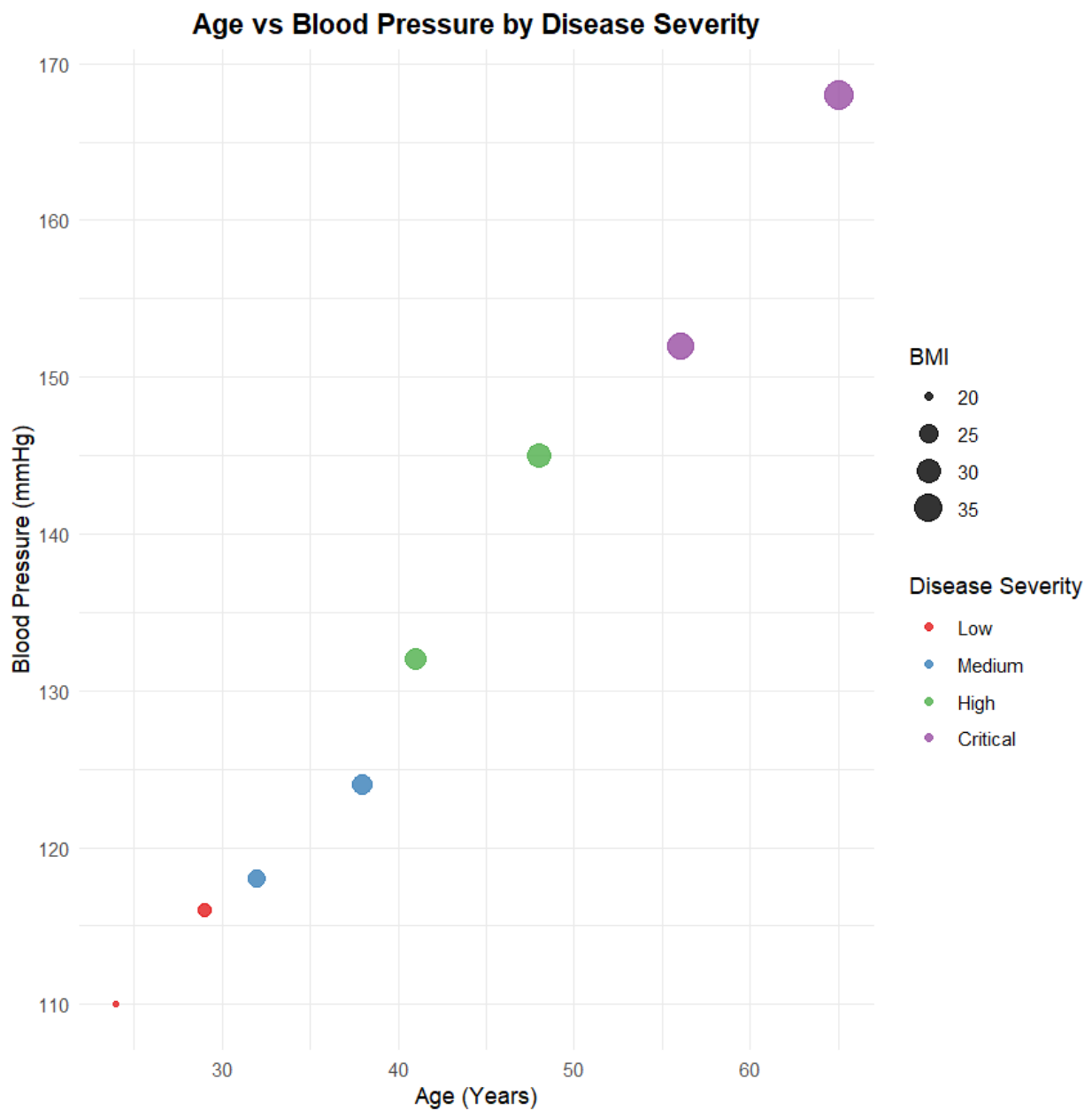
Each question combines **Data Visualization + R Programming + another discipline** to promote interdisciplinary learning.

Question 1 (Healthcare Analytics)

Dataset

Patient	Age	BMI	Blood Pressure	Disease Severity
P1	24	19.8	110	Low
P2	32	24.5	118	Medium
P3	41	28.1	132	High
P4	48	30.3	145	High
P5	56	33.2	152	Critical
P6	65	36.5	168	Critical
P7	38	26.2	124	Medium
P8	29	22.1	116	Low

```
1 library(ggplot2)
2 health_data <- data.frame(
3   Patient = c("P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8"),
4   Age = c(24, 32, 41, 48, 56, 65, 38, 29),
5   BMI = c(19.8, 24.5, 28.1, 30.3, 33.2, 36.5, 26.2, 22.1),
6   BloodPressure = c(110, 118, 132, 145, 152, 168, 124, 116),
7   DiseaseSeverity = factor(
8     c("Low", "Medium", "High", "High", "Critical", "Critical", "Medium", "Low"),
9     levels = c("Low", "Medium", "High", "Critical")
10  )
11 )
12 ggplot(health_data,
13   aes(x = Age,
14     y = BloodPressure,
15     color = DiseaseSeverity,
16     size = BMI)) +
17   geom_point(alpha = 0.8) +
18   scale_color_brewer(palette = "Set1") +
19   labs(
20     title = "Age vs Blood Pressure by Disease Severity",
21     x = "Age (Years)",
22     y = "Blood Pressure (mmHg)",
23     color = "Disease Severity",
24     size = "BMI"
25   ) +
26   theme_minimal() +
27   theme(
28     plot.title = element_text(hjust = 0.5, face = "bold"),
29     legend.position = "right"
30   )
```



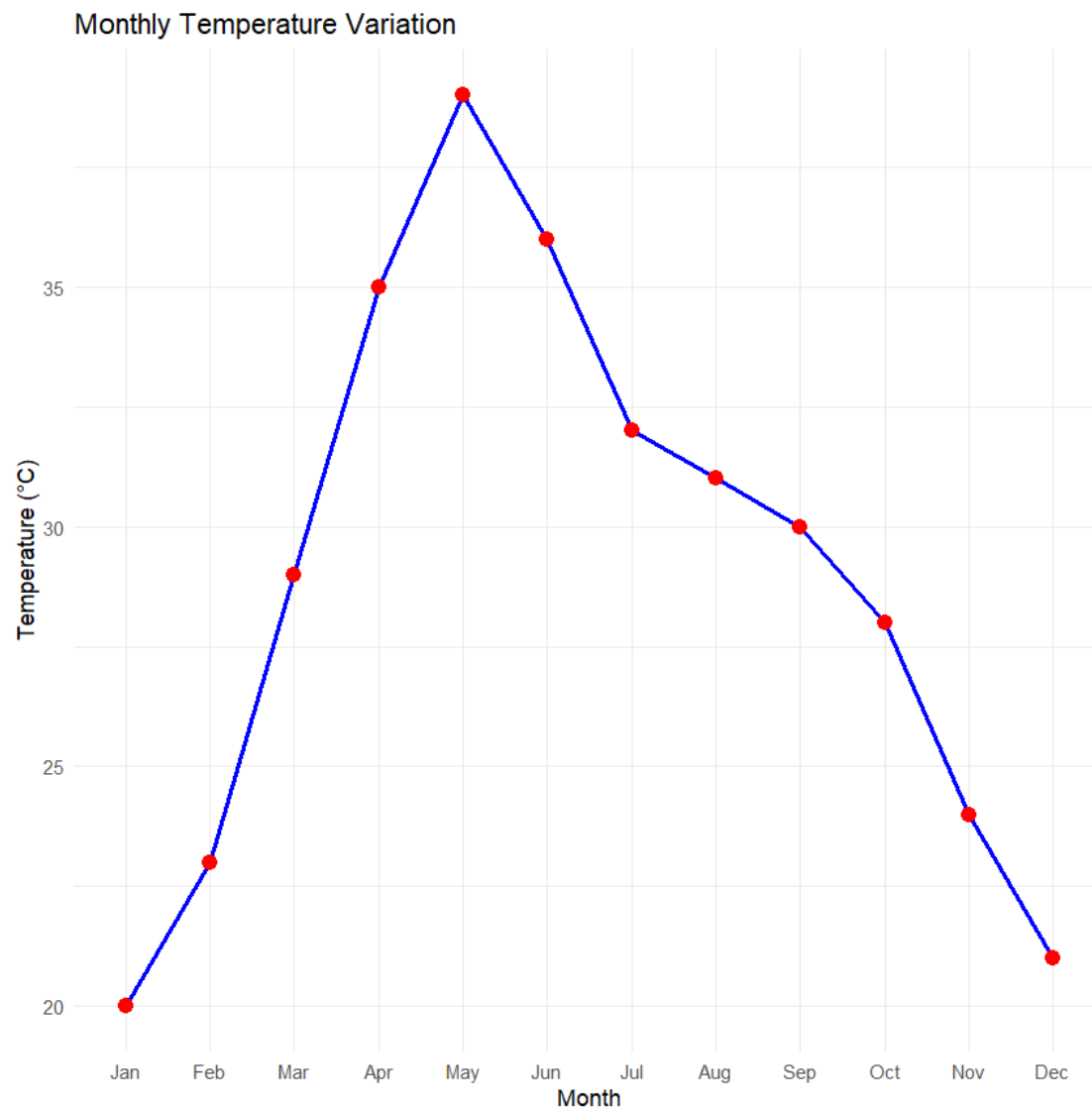
1. Blood pressure generally increases with age among the patients.
2. Patients with Critical disease severity have the highest blood pressure and BMI values.
3. Low severity patients are younger and have lower blood pressure.
4. Larger points indicate higher BMI, showing that patients with higher BMI tend to have higher blood pressure.
5. The color coding clearly distinguishes disease severity levels, making it easier to identify health risk patterns.

Question 2 (Climate Science)

Dataset

Month	Solar Radiation	Temperature (°C)	Rainfall (mm)
Jan	180	20	25
Feb	200	23	18
Mar	240	29	12
Apr	280	35	8
May	320	39	5
Jun	300	36	65
Jul	260	32	150
Aug	250	31	170
Sep	240	30	120
Oct	220	28	85
Nov	190	24	45
Dec	170	21	30

```
1 library(ggplot2)
2 climate <- data.frame(
3   Month = factor(c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
4                 levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov",
5                 Temperature = c(20, 23, 29, 35, 39, 36, 32, 31, 30, 28, 24, 21)
6 )
7 ggplot(climate, aes(x = Month, y = Temperature, group = 1)) +
8   geom_line(color = "blue", linewidth = 1) +
9   geom_point(color = "red", size = 3) +
10   labs(
11     title = "Monthly Temperature Variation",
12     x = "Month",
13     y = "Temperature (°C)"
14   ) +
15   theme_minimal()
```

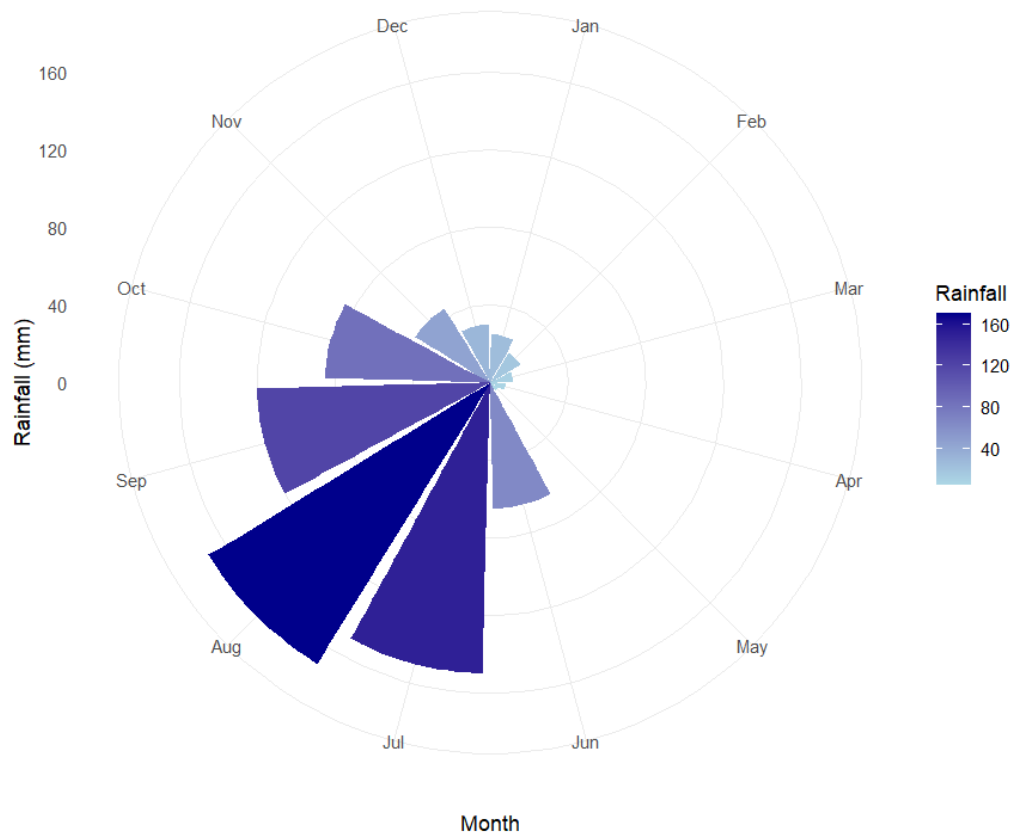


```

1 library(ggplot2)
2 climate_data <- data.frame(
3   Month = factor(c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
4     "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
5     levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
6       "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")),
7   SolarRadiation = c(180, 200, 240, 280, 320, 300, 260, 250, 240, 220, 190, 170),
8   Temperature = c(20, 23, 29, 35, 39, 36, 32, 31, 30, 28, 24, 21),
9   Rainfall = c(25, 18, 12, 8, 5, 65, 150, 170, 120, 85, 45, 30)
10 )
11 ggplot(climate_data, aes(x = Month, y = Rainfall, fill = Rainfall)) +
12   geom_col() +
13   coord_polar() +
14   labs(
15     title = "Annual Rainfall (Polar Coordinate Chart)",
16     x = "Month",
17     y = "Rainfall (mm)"
18   ) +
19   scale_fill_gradient(low = "lightblue", high = "darkblue") +
20   theme_minimal()

```

Annual Rainfall (Polar Coordinate Chart)



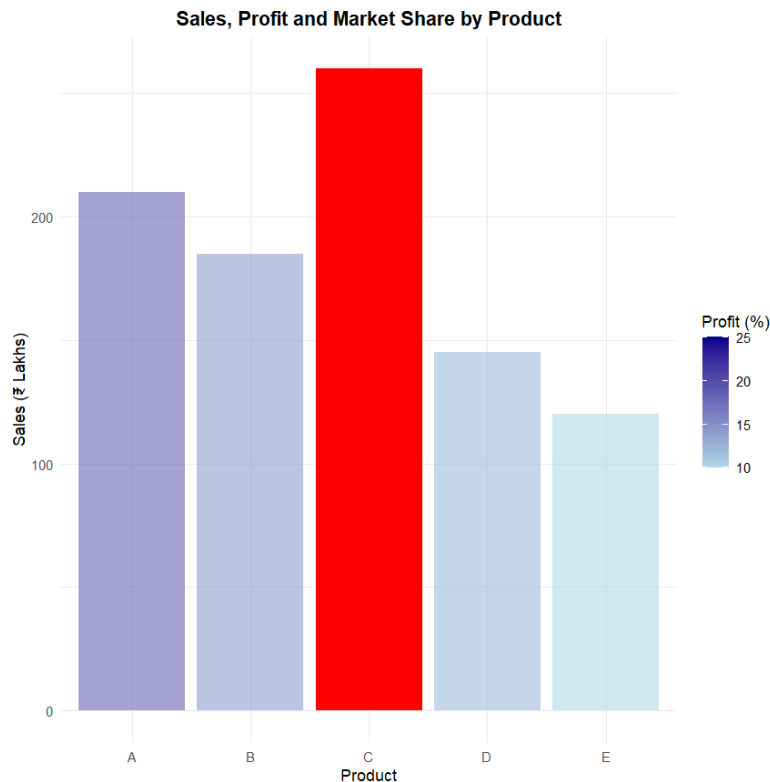
The Cartesian coordinate system is more suitable for analyzing and comparing exact climate measurements such as monthly temperature trends because it provides a clear representation of changes over time. In contrast, the Polar coordinate system is better for visualizing seasonal or annual climate cycles, such as rainfall patterns, by highlighting the cyclical nature of the data. Using both coordinate systems together provides a more comprehensive understanding of climate data by combining accurate trend analysis with seasonal pattern recognition.

Question 3 (Business Analytics & Marketing)

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Market Share (%)
A	210	18	24
B	185	14	20
C	260	25	28
D	145	12	15
E	120	10	13

```
1 library(ggplot2)
2 business_data <- data.frame(
3   Product = c("A", "B", "C", "D", "E"),
4   Sales = c(210, 185, 260, 145, 120),
5   Profit = c(18, 14, 25, 12, 10),
6   MarketShare = c(24, 20, 28, 15, 13)
7 )
8 business_data$Highlight <- ifelse(
9   business_data$MarketShare == max(business_data$MarketShare),
10  "Highest Market Share",
11  "Other Products"
12 )
13 ggplot(business_data,
14        aes(x = Product,
15            y = Sales,
16            fill = Profit,
17            alpha = Highlight)) +
18  geom_col() +
19  scale_fill_gradient(low = "lightblue", high = "darkblue") +
20  scale_alpha_manual(values = c("Highest Market Share" = 1,
21                                "Other Products" = 0.6),
22                    guide = "none") +
23  geom_col(
24    data = subset(business_data, Highlight == "Highest Market Share"),
25    fill = "red"
26 ) +
27  labs(
28    title = "Sales, Profit and Market Share by Product",
29    x = "Product",
30    y = "Sales (₹ Lakhs)",
31    fill = "Profit (%)"
32 ) +
33  theme_minimal() +
34  theme(
35    plot.title = element_text(hjust = 0.5, face = "bold")
36 )
```



The chosen color palette improves business decision-making in the following ways:

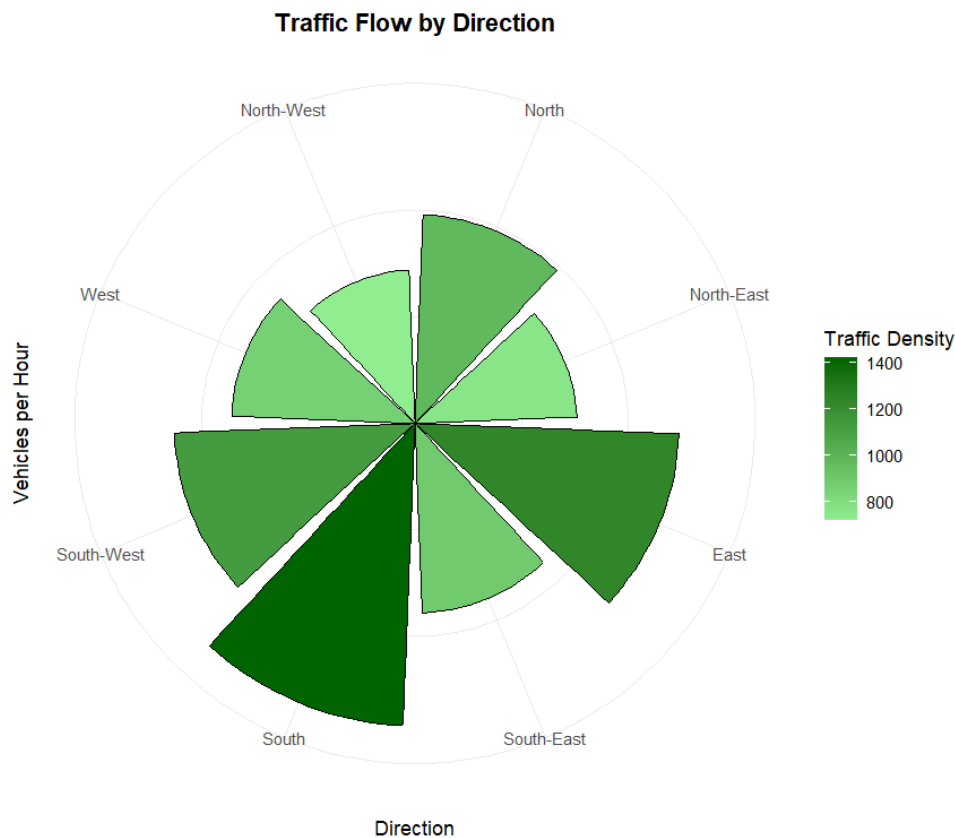
1. The sequential color palette (light blue to dark blue) clearly represents increasing profit percentages, making it easy to identify more profitable products at a glance.
2. Using darker shades for higher profits helps managers quickly recognize top-performing products without reading the exact values.
3. Highlighting the product with the highest market share in a contrasting red color immediately draws attention to the market leader, supporting strategic decisions.
4. The combination of bar height (sales), color intensity (profit), and contrasting highlight (market share) allows multiple business metrics to be analyzed in a single visualization.
5. The color scheme reduces visual clutter and improves readability, enabling faster and more accurate comparisons between products.
6. Overall, the palette supports informed business decisions by making important performance trends and standout products easy to identify.

Question 4 (Transportation Engineering)

Dataset

Direction	Vehicles/hour
North	980
North-East	760
East	1240
South-East	890
South	1420
South-West	1130
West	860
Direction	Vehicles/hour
North-West	720

```
1 library(ggplot2)
2 traffic_data <- data.frame(
3   Direction = factor(
4     c("North", "North-East", "East", "South-East",
5       "South", "South-West", "West", "North-West"),
6     levels = c("North", "North-East", "East", "South-East",
7               "South", "South-West", "West", "North-West")
8   ),
9   Vehicles = c(980, 760, 1240, 890, 1420, 1130, 860, 720)
10 )
11 ggplot(traffic_data,
12       aes(x = Direction,
13           y = Vehicles,
14           fill = Vehicles)) +
15   geom_col(color = "black") +
16   coord_polar() +
17   scale_fill_gradient(low = "lightgreen", high = "darkgreen") +
18   labs(
19     title = "Traffic Flow by Direction",
20     x = "Direction",
21     y = "Vehicles per Hour",
22     fill = "Traffic Density"
23   ) +
24   theme_minimal() +
25   theme(
26     plot.title = element_text(hjust = 0.5, face = "bold"),
27     axis.text.y = element_blank(),
28     axis.ticks = element_blank()
29   )
```



A Polar chart is more suitable than a Cartesian bar chart for directional traffic analysis for the following reasons:

1. Represents compass directions naturally: A polar chart arranges categories in a circular layout that matches real-world compass directions (North, East, South, West, etc.), making the visualization intuitive.
2. Highlights directional patterns: It clearly shows how traffic is distributed around an intersection, allowing users to quickly identify the busiest and least busy directions.
3. Better for circular data: Traffic flow by direction is cyclical rather than linear. A polar chart preserves this relationship, while a Cartesian bar chart displays directions as a straight sequence, which does not reflect their actual spatial arrangement.
4. Improves traffic planning: Engineers can easily identify congestion hotspots and compare traffic from opposite or adjacent directions, helping with signal timing, road design, and traffic management.
5. Provides an intuitive visual overview: Although a Cartesian bar chart is better for precise numerical comparisons, a polar chart offers a clearer representation of directional traffic flow, making it more effective for analyzing movement patterns at intersections and roundabouts.

Question 5 (Environmental Sustainability Dashboard)

Dataset

City	AQI	Green Cover (%)	CO ₂ Emission	Population (Lakhs)
A	48	45	1.8	9
B	82	32	3.5	15
C	135	21	5.6	22
D	165	18	7.4	28
E	72	38	2.9	12
F	95	30	4.2	18

```
1 library(ggplot2)
2 library(gridExtra)
3 env_data <- data.frame(
4   City = c("A", "B", "C", "D", "E", "F"),
5   AQI = c(48, 82, 135, 165, 72, 95),
6   GreenCover = c(45, 32, 21, 18, 38, 30),
7   CO2 = c(1.8, 3.5, 5.6, 7.4, 2.9, 4.2),
8   Population = c(9, 15, 22, 28, 12, 18)
9 )
10 env_data$AQI_Category <- cut(
11   env_data$AQI,
12   breaks = c(0, 50, 100, 150, 200),
13   labels = c("Good", "Moderate", "Poor", "Very Poor"))
14 p1 <- ggplot(env_data,
15   aes(x = GreenCover,
16     y = AQI,
17     color = AQI_Category,
18     size = Population)) +
19   geom_point(alpha = 0.8) +
20   scale_color_manual(values = c("Good" = "green", "Moderate" = "yellow",
21     "Poor" = "orange", "Very Poor" = "red")) +
22   labs(
23     title = "AQI vs Green Cover",
24     x = "Green Cover (%)",
25     y = "AQI",
26     color = "AQI Category",
27     size = "Population (Lakhs)"
28   ) +
29   theme_minimal()
30 p2 <- ggplot(env_data,
31   aes(x = City,
32     y = CO2,
33     fill = AQI_Category)) +
34   geom_col() +
35   scale_fill_manual(values = c("Good" = "green", "Moderate" = "yellow",
36     "Poor" = "orange", "Very Poor" = "red")) +
37   labs(
38     title = "CO2 Emissions by City",
39     x = "City",
40     y = "CO2 Emissions",
41     fill = "AQI Category"
42   ) +
43   theme_minimal()
```

